

CENG322 - PA1 Report

Ahmet Kurt
290201034

March 21, 2025

1. Introduction

This report explains the steps that I take to execute PolyBench-C benchmarks on the UHEM system. The purpose of this assignment is to:

- Compile and run five PolyBench-C benchmarks.
- Submit and execute the programs on UHEM.
- Collect execution times over 10 runs.
- Process and compute average execution times.

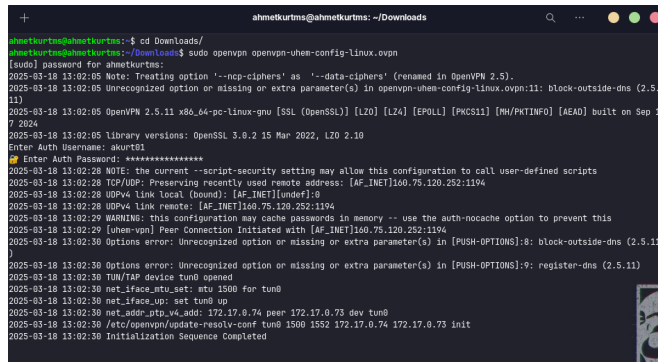
2. Selected Benchmarks

The following benchmarks were selected:

- **2mm** (Linear Algebra - Kernels)
- **3mm** (Linear Algebra - Kernels)
- **atax** (Linear Algebra - Kernels)
- **bicg** (Linear Algebra - Kernels)
- **doitgen** (Linear Algebra - Kernels)

3. Connecting the VPN

Firstly I downloaded the file named `openvpn-uhem-config-linux.ovpn` with my username and password. I move to the directory and run it with command `sudo openvpn`.



```
ahmetkurtms@ahmetkurtms: ~/Downloads
ahmetkurtms@ahmetkurtms:~$ cd Downloads/
ahmetkurtms@ahmetkurtms:~/Downloads$ sudo openvpn openvpn-uhem-config-linux.ovpn
[sudo] password for ahmetkurtms:
2025-03-18 13:02:05 Note: Treating option '--data-ciphers' as '--data-ciphers' (renamed in OpenVPN 2.5).
2025-03-18 13:02:05 Unrecognized option or missing or extra parameter(s) in openvpn-uhem-config-linux.ovpn:11: block-outside-dns (2.5.11)
2025-03-18 13:02:05 OpenVPN 2.5.11 x86_64-pc-linux-gnu [SSL (OpenSSL)] [LZO] [LZ4] [EPOLL] [PKCS11] [MH/PTHINFO] [AEAD] built on Sep 17 2024
2025-03-18 13:02:05 Library versions: OpenSSL 3.0.2 15 Mar 2022, LZO 2.10
Enter Auth Username: akurt81
Enter Auth Password: *****
2025-03-18 13:02:28 NOTE: the current --script-security setting may allow this configuration to call user-defined scripts
2025-03-18 13:02:28 TCP/UDP: Preserving recently used remote address: [AF_INET]160.75.120.252:1194
2025-03-18 13:02:28 UDPv4 link local (bound): [AF_INET]undef:0
2025-03-18 13:02:28 UDPv4 link remote: [AF_INET]160.75.120.252:1194
2025-03-18 13:02:29 WARNING: this configuration may cache passwords in memory -- use the auth-nocache option to prevent this
2025-03-18 13:02:29 [uhem-vpn] Peer Connection Initiated with [AF_INET]160.75.120.252:1194
2025-03-18 13:02:30 Options error: Unrecognized option or missing or extra parameter(s) in [PUSH-OPTIONS]:8: block-outside-dns (2.5.11)
2025-03-18 13:02:30 Options error: Unrecognized option or missing or extra parameter(s) in [PUSH-OPTIONS]:9: register-dns (2.5.11)
2025-03-18 13:02:30 TUN/TAP device tun0 opened
2025-03-18 13:02:30 net_iface_atu_sat: mtu 1500 for tun0
2025-03-18 13:02:30 net_iface_up: set tun0 up
2025-03-18 13:02:30 net_addr_ptp_v4_add: 172.17.0.74 peer 172.17.0.73 dev tun0
2025-03-18 13:02:30 /etc/openvpn/update-resolv-conf tun0 1500 1552 172.17.0.74 172.17.0.73 init
2025-03-18 13:02:30 Initialization Sequence Completed
```

Figure 1: Connecting the VPN

4. Connecting to the UHEM System

After the VPN connection with command: `ssh username@10.128.2.32` and with my password, connected to the system.



```
ahmetkurtms@ahmetkurtms: ~/Downloads
ahmetkurtms@ahmetkurtms:~/Downloads$ ssh akurt@10.128.2.32
akurt@10.128.2.32's password:
Last login: Tue Mar 18 11:46:03 2025 from 10.128.2.240

=====
Kullanıcı komutları ile hızlıca komut öğrenebilirsiniz.
Use komutlar command for a list of frequently used commands.

=====
UHEM disk sistemlerinin yedeklenmesi yapılmamaktadır, bu nedenle
önemli verilerinizi lütfen UHEM dışında da yedekleyin.

=====
Sarıyer kumesi, disk sistemindeki doluluk oranı %98'e ulaştığından,
diskte yeterli boş alan sağlanana dek yeni işlere kapatılmıştır.

Kullanıcılara:

- gereksizsin diyemadıkları geçici dosyaları silmeleri
- doluluk nedeniyle ortaya çıkabilecek bir kesintiye karşı
  kullanıcılara ön verilerini yedeklemeleri
- ilerideki çalışmalar için yazılım ve donanım acısından
  güncel kumemiz olan Atatürk'ü değerlendirmeleri
  önerilmektedir.

=====
Elektrik altyapısında gerçekleştirilecek çalışma nedeniyle,
29 Mart 2025 sabahı tüm sistemleriniz kapatılacaktır.

Lütfen planlarınızı yaparken o anda çalışan işlerinizin
olumsuzuna diyet edinir. Sistemler kapatılınca çalışan
işlerin olası durumunda işler sonlandırılacak ve herhangi
bir hesaplama zamanı ladesi yapılmayacaktır.

Kesinti sırasında web ve eposta dahil hiç bir sunucuya
elektrik verilemeyeceği için hiç bir sunucuya erişim
olmayacaktır.

Anlayışınız için teşekkür ederiz.

=====
```

Figure 2: Connecting the UHEM

5. Downloading the Polybench C Benchmark

With the command `wget` and `tar`, I downloaded the link which given in the homework PDF file. And extract it in the system.



```
ahmetkurtms@ahmetkurtms: ~/Downloads
[akurt@10.128.2.32 ~]$ wget https://www.cs.colostate.edu/~pouchet/software/polybench/download/polybench-c-3.2.tar.gz
--2025-03-18 13:55:47-- https://www.cs.colostate.edu/~pouchet/software/polybench/download/polybench-c-3.2.tar.gz
Resolving www.cs.colostate.edu (www.cs.colostate.edu)... 129.82.45.48
Connecting to www.cs.colostate.edu (www.cs.colostate.edu)|129.82.45.48|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 22993 (22K) [application/x-gzip]
Saving to: 'polybench-c-3.2.tar.gz'

100%[*****] 22,993 143KB/s in 6.2s

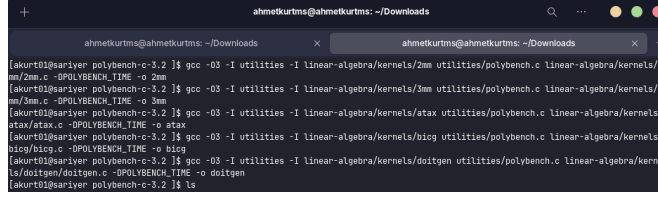
2025-03-18 13:55:48 (141 KB/s) - 'polybench-c-3.2.tar.gz' saved [22993/22993]

[akurt@10.128.2.32 ~]$ tar xvfz
polybench-c-3.2.tar.gz
./
./bash_history
./cache/
[akurt@10.128.2.32 ~]$ tar xvfz
polybench-c-3.2.tar.gz
./
./bash_history
./cache/
[akurt@10.128.2.32 ~]$ tar xvfz polybench-c-3.2.tar.gz
polybench-c-3.2/
polybench-c-3.2/.ssh/
```

Figure 3: Download-Extract Benchmark

6. Compilation of Benchmarks

After navigating to the directory, I compiled the benchmarks using the `gcc` command with optimization level `-O3`. Here are the commands I used:



```
ahmetkurtms@ahmetkurtms: ~/Downloads
[skurt@ariyer polybench-c-3.2]$ gcc -O3 -I utilities -I linear-algebra/kernels/2mm utilities/polybench.c linear-algebra/kernels/2mm/2mm.c -DPOLYBENCH_TIME -o 2mm
[skurt@ariyer polybench-c-3.2]$ gcc -O3 -I utilities -I linear-algebra/kernels/3mm utilities/polybench.c linear-algebra/kernels/3mm/3mm.c -DPOLYBENCH_TIME -o 3mm
[skurt@ariyer polybench-c-3.2]$ gcc -O3 -I utilities -I linear-algebra/kernels/atax utilities/polybench.c linear-algebra/kernels/atax/atax.c -DPOLYBENCH_TIME -o atax
[skurt@ariyer polybench-c-3.2]$ gcc -O3 -I utilities -I linear-algebra/kernels/bicg utilities/polybench.c linear-algebra/kernels/bicg/bicg.c -DPOLYBENCH_TIME -o bicg
[skurt@ariyer polybench-c-3.2]$ gcc -O3 -I utilities -I linear-algebra/kernels/doitgen utilities/polybench.c linear-algebra/kernels/doitgen/doitgen.c -DPOLYBENCH_TIME -o doitgen
[skurt@ariyer polybench-c-3.2]$ ls
```

Figure 4: Compiling Benchmarks

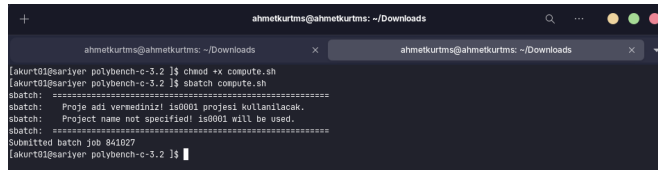
7. compute.sh

To automate execution, I used the `compute.sh` script. It runs each benchmark 10 times, measures execution time, and stores results in `output_shell.txt`.



```
#!/bin/bash
#benchmarks that I select
BENCHMARKS=("2mm" "3mm" "atax" "bicg" "doitgen")
#output
OUTPUT="output_shell.txt"
echo "Benchmark Execution Times" > $OUTPUT
#10 times run
for benchmark in ${BENCHMARKS[@]}; do
    echo "Running the $benchmark" >> $OUTPUT #output to the file
    for i in {1..10}; do
        { time ./$benchmark; } 2>> $OUTPUT #time command
    done
done
echo "Execution finished."
```

Figure 5: compute.sh



```
ahmetkurtms@ahmetkurtms: ~/Downloads
[skurt@ariyer polybench-c-3.2]$ chmod +x compute.sh
[skurt@ariyer polybench-c-3.2]$ sbatch compute.sh
sbatch: =====
sbatch:  Proj: adi vamedirizl is0001 proj01 kollanilacak.
sbatch:  Project name not specified! is0001 will be used.
sbatch: =====
Submitted batch job 841027
[skurt@ariyer polybench-c-3.2]$
```

Figure 6: Run the script

After that, it select the project name automatically. I waited for the run and the slurm - output files exist.

8. Output_shell.txt

```
1 Running 2mm
2 real 5.47
3 real 5.47
4 real 5.48
5 real 5.47
6 real 5.47
7 real 5.47
8 real 5.48
9 real 5.47
10 real 5.47
11 real 5.44
12 Running 3mm
13 real 8.21
14 real 8.19
15 real 8.20
16 real 8.19
17 real 8.19
18 real 8.19
19 real 8.19
20 real 8.19
21 real 8.20
22 real 8.19
23 Running atax
24 real 0.10
25 real 0.08
26 real 0.08
27 real 0.08
28 real 0.08
29 real 0.08
30 real 0.08
31 real 0.08
32 real 0.08
33 real 0.08
34 Running bicg
35 real 0.12
36 real 0.10
37 real 0.10
38 real 0.10
39 real 0.10
40 real 0.10
41 real 0.10
42 real 0.10
43 real 0.10
44 real 0.10
45 Running doitgen
46 real 0.37
47 real 0.36
48 real 0.36
49 real 0.36
50 real 0.36
51 real 0.36
52 real 0.36
53 real 0.36
54 real 0.36
55 real 0.36
```

9. process.sh

After execution, ‘process.sh’ extracts execution times from ‘outputshell.txt’ and computes the average for each benchmark.

```
1  #!/bin/bash
2
3  INPUT="output_shell.txt"
4  OUTPUT="output_times.txt"
5
6  > $OUTPUT # Clear previous data
7
8  benchmark_name=""
9  total=0
10 entries=0
11
12 while IFS= read -r line; do
13     if [[ $line == "Running "+* ]]; then
14         # If there are pre-entries, calc. and write avg.
15         if [[ $entries -gt 0 ]]; then
16             average_time=$(awk "BEGIN {print $total / $entries}")
17             echo "Benchmark: $benchmark_name - Avg Time: $average_time sec" >> $OUTPUT
18         fi
19         benchmark_name=$(echo $line | cut -d' ' -f2)
20         total=0
21         entries=0
22     elif [[ $line == real* ]]; then
23         time_value=$(echo $line | awk '{print $2}')
24         total=$(awk "BEGIN {print $total + $time_value}")
25         entries=$((entries + 1))
26     fi
27 done < "$INPUT"
28
29 if [[ $entries -gt 0 ]]; then
30     average_time=$(awk "BEGIN {print $total / $entries}")
31     echo "Benchmark: $benchmark_name - Avg Time: $average_time sec" >> $OUTPUT
32 fi
33
34 echo "All average execution times saved as $OUTPUT."
```

Figure 7: process.sh

Running the process.sh and the output:

```
ahmetkurtms@ahmetkurtms: ~/Downloads
ahmetkurtms@ahmetkurtms: ~/Downloads
[akurt@ahmetkurtms polybench-c-3.2] $ chmod +x process.sh
[akurt@ahmetkurtms polybench-c-3.2] $ ./process.sh
All average execution times saved as output_times.txt.
[akurt@ahmetkurtms polybench-c-3.2] $ cat output_times.txt
Benchmark: 2mm - Avg Time: 5.469 sec
Benchmark: 3mm - Avg Time: 8.194 sec
Benchmark: atax - Avg Time: 0.082 sec
Benchmark: bicg - Avg Time: 0.102 sec
Benchmark: doitgen - Avg Time: 0.361 sec
[akurt@ahmetkurtms polybench-c-3.2] $
```

Figure 8: process.sh run

10. Results

The final execution times are stored in ‘output_times.txt’:

```
1 Benchmark: 2mm - Avg Time: 5.469 sec
2 Benchmark: 3mm - Avg Time: 8.194 sec
3 Benchmark: atax - Avg Time: 0.082 sec
4 Benchmark: bicg - Avg Time: 0.102 sec
5 Benchmark: doitgen - Avg Time: 0.361 sec
```

11. Commands

Below is a summary of all the commands used throughout the process:

Step	Command
VPN Connection	<code>sudo openvpn --config openvpn-uhem-config-linux.ovpn</code>
SSH Connection to UHeM	<code>ssh username@10.128.2.32</code>
Download PolyBench-C	<code>wget <URL></code>
Extract the Benchmark Suite	<code>tar -xvzf polybench-c-3.2.tar.gz</code>
Navigate to PolyBench Directory	<code>cd polybench-c-3.2</code>
Compile Benchmarks	<pre>1 gcc -O3 -I utilities -I -path- \ 2 utilities/polybench.c -c-path \ 3 -DPOLYBENCH_TIME -o benchmark-name</pre>
Make compute.sh Executable	<code>chmod +x compute.sh</code>
Submit SLURM Job	<code>sbatch slurm_sample.sh</code>
Monitor SLURM Job	<code>squeue -u \$USER</code>
Check SLURM Output	<code>cat slurm-XXXXX.out</code>
Check Benchmark Execution Output	<code>cat output_shell.txt</code>
Make process.sh Executable	<code>chmod +x process.sh</code>
Run process.sh	<code>./process.sh</code>
View Final Execution Times	<code>cat output_times.txt</code>

Table 1: Commands Used Throughout the Process

12. Conclusion

In this assignment, I successfully executed and analyzed PolyBench-C benchmarks on the UHEM system. Throughout the process, I learned how to compile, execute, and monitor benchmark programs. The main steps I completed:

- Connecting to the UHEM system via VPN and SSH.
- Downloading and extracting the PolyBench-C benchmark.
- Compiling five selected benchmarks: `2mm`, `3mm`, `atax`, `bicg`, and `doitgen`.
- Writing and executing the `compute.sh` script, which ran each benchmark 10 times.
- Writing and executing the `process.sh` script to calculate the average execution time of each benchmark.